

Problem Set 9: Digital & Platform Markets

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before Lecture 1 of Module 10]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with $\langle AI \rangle$ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Two-sided pricing with numbers — 15 points). *A rideshare platform faces rider demand $R(p) = 100 - 20p$ (riders per day at rider fee p) and incurs a cost of \$3 per rider served.*

- (a) **(4 pts)** *Treating the rider side in isolation, find the profit-maximizing fee and the resulting profit. Hint: the profit function is a parabola; its peak is midway between its zeros.*
- (b) **(5 pts)** *Now recognize the platform's other side: each active rider generates \$3 of profit on the driver side (higher utilization, commissions). The \$3 service cost still applies; the driver-side \$3 is additional profit per served rider. Find the new profit-maximizing rider fee and total platform profit (rider-side margin plus driver-side profit). State plainly what is remarkable about the optimal fee relative to the \$3 cost.*
- (c) **(3 pts)** *A competition authority calls the fee in (b) predatory pricing. Write the platform's two-sentence defense using your arithmetic, and one sentence on what evidence would distinguish genuine predation from two-sided pricing.*
- (d) **(3 pts)** *In one paragraph: which side of a two-sided platform should be the "money side"? Use the elasticity-and-externality rule of thumb and one real example (cards, media, delivery apps).*

Question 2 (Interference in a platform A/B test — 15 points). *A rideshare market has fixed driver capacity of 1,000 completed rides per day. Baseline: 1,000 ride requests per day, so all requests are served; riders are split into two equal halves for an experiment (500 baseline requests each). The treated half receives a discount that raises its requests by 20%; the control half's requests are unchanged. When total requests exceed capacity, completed rides are allocated proportionally to requests.*

- (a) **(5 pts)** *Compute each group's completed rides under the experiment and the naive treatment estimate (treated minus control completed rides, and in percent of the 500-ride baseline).*
- (b) **(4 pts)** *Now compute the all-treated counterfactual: every rider gets the discount. What happens to completed rides relative to baseline, and what is the true effect of full rollout on completed rides?*

- (c) (3 pts) Name the assumption that failed, say exactly where it failed (which resource do the groups share), and state the direction of the naive bias in one sentence.
- (d) (3 pts) Two design fixes: cluster randomization and switchback randomization. One sentence each: what the fix buys, and the price it pays.

Part B: Experimental Design (45 points)

Question 3 (Testing a marketplace feature under interference). A large e-commerce marketplace wants to test a “verified seller” badge: sellers who pass a quality audit get a visible badge on their listings. The concern: badges may not grow the market — they may merely reallocate demand from unbadged to badged sellers. Design the experiment that separates market growth from reallocation. Specify:

- (a) (6 pts) The setting and stakes: the marketplace structure (categories, sellers per category, buyer search behavior), why within-category demand stealing is the expected interference channel, and why the growth-vs-reallocation question determines whether the feature should ship.
- (b) (12 pts) Treatments and randomization: your design must include a randomization unit chosen FOR the interference channel (justify cluster vs. switchback vs. a mixed design), and must identify (1) the total effect on badged sellers, (2) the spillover onto unbadged sellers in the same market, and (3) the market-level net effect. State precisely which comparison delivers each, and what a naive seller-level randomization would have concluded.
- (c) (9 pts) Measurement: primary outcomes at seller and market level, guardrail metrics (Kohavi-style), novelty-effect handling (how long you run and why), and the sample-ratio-mismatch check.
- (d) (9 pts) Hypotheses and power: state the growth-vs-reallocation hypotheses quantitatively (e.g. badged sellers +15%, unbadged $-x\%$, market net $+y\%$), and give the cluster-level power reasoning: number of category-markets per arm given a stated intra-cluster correlation, and which comparison (the spillover or the net effect) drives the sample size and why.
- (e) (9 pts) The biggest threat to YOUR design and your answer to it. (Candidates: buyers search across categories, so clusters leak; badge supply is endogenous — sellers select into auditing; switchback carryover through reviews and repeat purchase.)

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The sellers that colluded in three rounds). You run `module9-simulation-lab.ipynb`. (For reference: the lab’s prompts hand each agent a plain-language description of demand and costs; *history* agents see the full price/profit path each round, *amnesia* agents are stateless; and the lab records only prices — no reasoning traces.) In the history treatment, your LLM seller pairs converge to the joint-monopoly price of 6 within three rounds and hold it there, with no communication. Calvano et al.’s (2020) Q-learning agents needed on the order of 100,000 learning episodes to reach supra-competitive prices. Your lab partner drafts an abstract: “LLM pricing agents collude faster than reinforcement learners — deployed LLM pricing tools pose an immediate collusion risk.”

- (a) (8 pts) Explain why three-round convergence to exactly the monopoly price is NOT evidence of “learning to collude” in Calvano’s sense, and give two mechanisms (at least one LLM-specific)

that could produce it. Hint: what did your agents receive in the prompt that Calvano's Q-learners had to discover through experience — and what has your model read?

- (b) **(9 pts)** Design the diagnostic battery: your design must include (i) the hidden-demand run (agents observe only their own realized sales, never the demand function — the information condition of deployed pricing tools) and (ii) the manual-deviation probe (force one agent to cut price for one round; Calvano's signature is punishment-then-forgiveness). Add at least one more (shifted cost/benchmark numbers; three-seller markets). State the predicted result pattern under each mechanism from (a). Run it if API budget allows and report; otherwise state predictions precisely.
- (c) **(8 pts)** The twist (Module 6's lesson returns): argue that a properly designed version of this simulation IS a legitimate, policy-relevant study — of what, exactly? State the research question it answers, the one design condition that must hold for policy relevance (hint: information), and the one-sentence scope limitation for the abstract your partner should write instead.

Attach your AI transcripts and, if you ran code, the results CSV.